Suggested FYP topics for 2026-27
by
Professor Sherman S. M. Chow
Email: smchow@ie.cuhk.edu.hk
Office: SHB 808

Project code :	SMC1
Project title :	Privacy-Preserving AI
No. of students :	2
Description of project :	Modern cryptography and modern AI now meet in systems that sound almost contradictory at first: running a transformer on private data without revealing the input, or proving that a large ML-style computation was carried out correctly without asking others to rerun it. This project starts from a real research artifact, pushing it one step further. You may choose one of the following directions. <u>Option A. Secure Transformer Inference</u> Secure inference means running a machine-learning model on private input data without revealing that data to the party performing the computation. SHAFT is an award-winning research artifact for secure transformer inference. This option extends the existing SHAFT codebase in a useful direction. Possible tasks: - add support for small new models or settings, - explore performance bottlenecks, - study accuracy, runtime, or communication tradeoffs, - improve reproducibility or ease of use, - adapt the system to one clearer use case, - or extend the comparison with other systems. https://www.ndss-symposium.org/ndss-paper/shaft-secure-handy-accurate-and-fast-transformer-inference https://github.com/andeskyl/SHAFT <u>Option B. zkSNARK for Verifiable Matrix Computation</u> A zkSNARK is a cryptographic proof system that lets one party prove that a computation was done correctly, while revealing very little beyond the validity of the statement. Start from Evalyn, a public artifact on proof-backed matrix computation with applications to verifiable deep learning. Possible directions include: - add a new matrix-computation workload, - benchmark performance under new settings, - study proof-size and runtime bottlenecks, - improve usability or reproducibility, - or adapt the framework to one small application. https://eprint.iacr.org/2025/1646 https://github.com/mirandaprivate/evalyn_asiacrypt A strong project could end with a working extension of a real research artifact, backed by a careful evaluation and a concrete result in private inference or verifiable ML.
Industrial collaboration :	No.

Project code :	SMC2
Project title :	Privacy-Preserving Fuzzy Matching
No. of students :	2
Description of project :	Many real systems need to match records that are similar but not exactly the same. Examples include name screening in payments, watchlist matching, and approximate set intersection. This project studies how to support such fuzzy matching while keeping sensitive data private. It can be approached from the application side or the cryptographic side, depending on your interest. Option A. Private fuzzy matching for payment/sanctions screening Study privacy-preserving name matching in a screening pipeline. Possible tasks: - choose one concrete compliance-style test, such as name / alias matching, noisy identifier matching, or duplicate-record detection, - construct or adapt a controlled dataset with realistic noise, spelling variation, or formatting changes, - define matching rule, distance notion, or threshold policy, - build a private matching pipeline for that rule, - compare the private version with a plaintext baseline, - add an extension, such as a second matching rule, a preprocessing layer, a batch mode, or a two-stage filtering design, - study how privacy, matching quality, and runtime change under different noise levels or rule choices. Option B. Fuzzy/Distance-aware private set intersection (PSI) Study a privacy-preserving tool for approximate set matching. Possible tasks: - choose one concrete similarity rule or distance notion, - implement one selected private approximate-matching design, - compare it with one exact-matching or simpler baseline, - add an extension, such as a second similarity rule, a preprocessing layer, a filtering stage, or a batched-query mode, - study how that choice changes runtime and matching behavior, - explain which parts of approximate matching become fundamentally harder than exact private set intersection. Possible expected outcome: - a clear problem statement, - a working prototype or reproducible experiment, - a careful evaluation Starter references: https://belfortlabs.com/blog/swift https://eprint.iacr.org/2026/441 https://www.usenix.org/conference/usenixsecurity25/presentation/legiest https://www.usenix.org/conference/usenixsecurity23/presentation/chakraborti-intersection
Industrial collaboration :	No.

Project code :	SMC3
Project title :	Multi-Client Searchable Encryption over MongoDB
No. of students :	2
Description of project :	This project studies how to support encrypted search for multiple users on top of a real database backend. Searchable encryption allows a server to store encrypted data and still answer limited search queries. Our recent breakthrough solves a two-decade open probelms of supporting multiple users while retaining sublinear efficiency and forward privacy properties. The goal is to build a prototype of anencrypted document store with multi-client access over a real document store such as MongoDB and study what works well, what leaks, and what costs appear in practice. Possible tasks: - design how encrypted documents and index entries are stored in MongoDB, - implement document insertion and exact keyword search, - support a selected reader/writer permission model, - benchmark storage overhead and query latency, - study the gap between asymptotic cost and observed cost. Possible expected outcome: - a working prototype over MongoDB or a similar backend, - a clear system design, - a careful benchmark and leakage discussion - a meaningful benchmark and leakage discussion, - a concrete database application that would have sounded unrealistic a few years ago. Starter references: https://staff.ie.cuhk.edu.hk/~smchow/4-enc.html https://www.ndss-symposium.org/ndss-paper/unus-pro-omnibus-multi-client-searchable-encryption-via-access-control https://www.usenix.org/conference/usenixsecurity22/presentation/wang-jiafan
Industrial collaboration :	No.

Project code :	SMC4
Project title :	Threshold Wallets and Robust Shared Control
No. of students :	2
Description of project :	This project studies how to control a shared wallet without giving full power to any one signer. A threshold wallet lets any *t* out of *n* signers approve a transaction. Our recent work studied robust threshold signing then improved efficiency. The goal is to build a threshold-wallet prototype and study how well it works under realistic conditions. Possible tasks: - implement a basic *t*-out-of-*n* approval flow, - add simple policies such as daily limits or approved recipients, - design a clear approval and audit interface, - study what happens when a signer is offline or delayed. Possible expected outcome: - a working wallet prototype, - a clear threat model, - an evaluation under realistic failure cases, - or a report explaining distributed trust, resilience, and usability in real-world scenarios Starter references: https://staff.ie.cuhk.edu.hk/~smchow/3-sig.html https://www.ndss-symposium.org/wp-content/uploads/2023-817-paper.pdf https://www.ndss-symposium.org/wp-content/uploads/2024-601-paper.pdf
Industrial collaboration :	No.

Project code :	SMC5
Project title :	Anonymous Credential Applications on Blockchain
No. of students :	2
Description of project :	Many blockchain systems would like to verify something about a user without exposing full identity, full history, or unnecessary linkage. Examples include good standing, reputation, membership, eligibility to access a service, or privacy-preserving transfers. Anonymous payment systems face a similar goal at the transaction level: support privacy without exposing unnecessary linkage information. This project studies privacy-preserving applications on blockchain using ideas from anonymous credentials and anonymous transaction systems. Recent works co-authored by our group cover scored credentials, anonymous reputation traits, and sustainability issues in anonymous blockchain systems. The goal is to build one concrete application and study, for example, what privacy or abuse risks remain. You may approach this from the credential side or from the private-transaction side. <u>Option A. Reputation and good-standing proofs</u> Build a blockchain-facing/decentralized system in which a user proves one reputation or status property without revealing full identity or full history. <u>Option B. Privacy-preserving transaction applications</u> Build an application in which users carry out a private action on chain with limited linkage. Examples: - a private payment or transfer flow, - a wallet interface for anonymous transactions, - a prototype service that accepts privacy-preserving payments, - an application that compares several transaction-privacy settings in terms of usability and long-term cost. Possible tasks: - implement an application flow on top of an existing non-anonymous design, - study how transaction parameters affect privacy and system cost Possible expected outcome:: - a clear problem statement, - a working prototype or reproducible artifact, - a careful evaluation, - and a report explaining limitations, lessons learned, etc. Starter references: https://staff.ie.cuhk.edu.hk/~smchow/2-blockchain.html https://staff.ie.cuhk.edu.hk/~smchow/3-sig.html https://staff.ie.cuhk.edu.hk/~smchow/scholar/sac.pdf https://eprint.iacr.org/2023/743
Industrial collaboration :	No.

Project code :	SMC6
Project title :	Privacy-Preserving Atomic Cross-Chain Exchange
No. of students :	2
Description of project :	Cross-chain exchange lets users swap assets across different blockchains without relying on a centralized exchange. Recent work studies how to make a cross-chain swap both atomic and privacy-preserving, so that either both sides complete the exchange or neither side does, while exposing less information than a fully transparent swap. This project starts from recent work on privacy-preserving atomic cross-chain exchange and asks how to turn that direction into a clearer user-facing application. Possible tasks: - build an application for private atomic exchange between one selected pair of testnets, - implement a clean user flow for posting, accepting, and completing a cross-chain swap, - compare a transparent baseline with a privacy-preserving swap flow in terms of latency, on-chain cost, and visible information, - add one application feature such as timeout handling, cancellation, status tracking, or basic audit logging. Starter reference: https://dl.acm.org/doi/10.1145/3708821.3736211
Industrial collaboration :	No.

Project code :	SMC7
Project title :	Privacy-Preserving AML and CBDC Payment Compliance
No. of students :	2
Description of project :	Modern payment systems must satisfy compliance requirements such as AML/CFT checks, sanctions screening, and transaction-policy enforcement. At the same time, many systems reveal more transaction data than is really necessary. This tension already appears in current official and industrial work. BIS Project Mandala studies compliance-by-design for cross-border payments, including compliance pre-validation and cryptographic proof of compliance. ECB material on the digital euro emphasizes strong privacy goals, including cash-like privacy for offline payments, while keeping AML checks with payment service providers during funding and defunding. This project studies how to build one privacy-preserving payment-compliance application. Possible tasks: - study what compliance checks can be made priva - implement a payment workflow with compliance checks, - build a wallet or middleware layer that reveals only the minimum output needed for approval, rejection, or escalation, - compare a full-disclosure baseline with a private design, - benchmark latency and throughput on synthetic payment data, Starter references: https://www.bis.org/about/bisih/topics/cbdc/mandala.htm https://www.ecb.europa.eu/euro/digital_euro/faqs/html/ecb.faq_digital_euro.en.html https://eprint.iacr.org/2024/1658
Industrial collaboration :	No.

Project code :	SMC8
Project title :	AI Security and Safety Applications
No. of students :	2
Description of project :	AI systems are now used in assistants, search, coding, content generation, and decision support. These systems create new security and safety problems such as prompt injection, unsafe tool use, jailbreaks, data leakage, and weak output controls. This project asks students to start from an existing AI application, model-serving pipeline, or public artifact and extend it in a certain direction. The focus is on careful implementation, evaluation, and analysis. Possible tasks: - build a small benchmark for selected attacks or failure modes, - implement one defense and compare it with baselines, - extend an existing open-source AI application with a security or safety layer, - study the tradeoff between safety control and task utility, - improve reproducibility, testing, or usability for a selected artifact. Possible expected outcome: - a clear problem statement, - a working prototype or reproducible artifact, - a careful evaluation, - and a report explaining limitations, lessons learned, etc.
Industrial collaboration :	No.

Project code :	SMC9
Project title :	Systems Security Applications
No. of students :	2
Description of project :	Many real systems must remain secure while still being usable, efficient, and deployable. Examples include authentication systems, access-control systems, sandboxing, software supply-chain defenses, browser or mobile security, network defenses, and secure system integration. This project asks students to start from an existing system, open-source codebase, or published artifact and extend it in one direction. The focus is on careful implementation, testing, and evaluation. The goal is not to invent a brand-new operating system or a full security architecture from scratch. Possible tasks: - harden one selected system component against a concrete threat, - build a proof-of-concept monitor or policy-enforcement layer, - reproduce a attack or defense experiment on an existing system, - compare a baseline system with one security-enhanced version, - improve testing, deployment, or reproducibility for a selected security artifact. Possible expected outcome: - a clear problem statement, - a working prototype or reproducible artifact, - a careful evaluation, - and a report explaining assumptions, limitations, and lessons learned.
Industrial collaboration :	No.